4 **(a)** State what is meant by an electric field.

.....

..... [1]

(b) Fig. 4.1 shows a small charged particle at a point A in a uniform electric field. The particle experiences an electric force F of 5.0×10^{-7} N. The grid lines are at intervals of 1.0 cm.

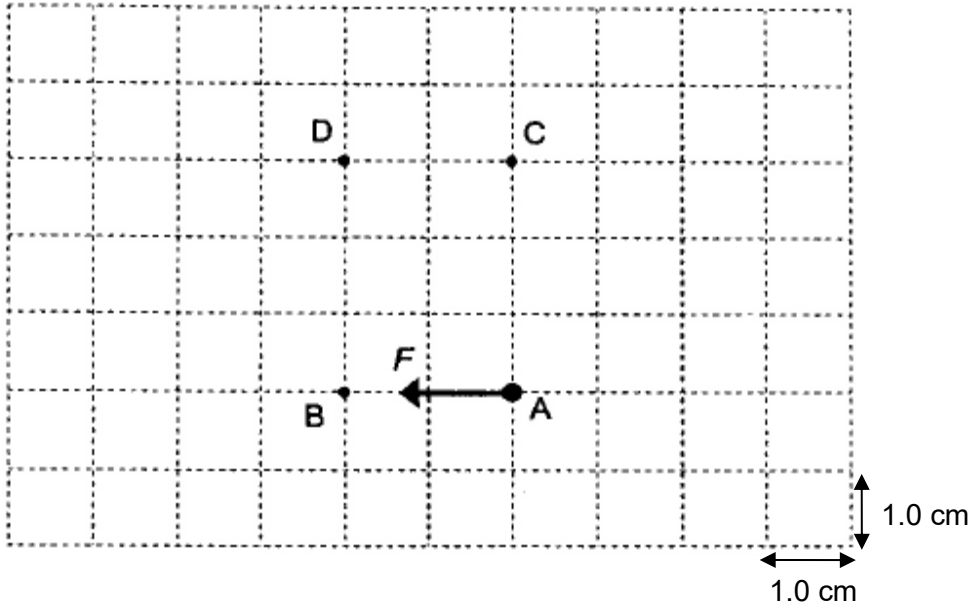

Fig. 4.1

Determine the work done by the electric force if the particle is moved

(i) from A to B,

W_{AB} = ... J [1]

(ii) from A to D.

W_{AD} = ... J [1]

(c) If the particle carries a charge of -2.5×10^{-11} C and point A is at a potential of +200 V, calculate the potential at point B due to the uniform field.

potential at B = ... V [2]

(d) The uniform electric field is produced by a pair of flat metal plates, one of which is earthed and the other is at a potential of +1000 V.

Sketch on Fig. 4.1

(i) the positions of the two plates. [2]

(ii) at least four equipotential lines in the region between the two plates. Label the value of the potential of each line that is drawn. [2]